

Researchers Use Lasers to Study Ancient Peruvian Mummies

For more than 5,000 years, humans have covered themselves with permanent artistic designs known as tattoos.

In a recent study, researchers used lasers to uncover highly complex designs of ancient tattoos on mummies from Peru.

The skin of the mummies and the black liquid, or ink, used to make the tattoos permitted researchers to observe fine details. These marks are not visible to the human eye, said study co-writer Michael Pittman of the Chinese University of Hong Kong.

The researchers examined around 100 mummies dating to around 775 years ago. The mummies were from coastal Peru's Chancay culture, which grew and developed before the Inca empire and the arrival of Europeans.

All the individuals had some form of tattoos on the back of their hands, knuckles, arms or other body parts. The study paid careful attention to four individuals with "exceptional tattoos" — designs of shapes such as triangles and diamonds, said Pittman.

This image provided by researchers shows intricate tattoo designs on a mummified human from Peru's Chancay culture in visible light and using laser-stimulated fluorescence. (Michael Pittman via AP)

It was not clear how the tattoos were created. But they are "of a quality that stands up against the really good electric tattooing of today," said Aaron Deter-Wolf of the Tennessee Division of Archaeology. Deter-Wolf was not involved in the research.

The study results appeared recently in the publication Proceedings of the National Academy of Sciences.

Using lasers that make skin glow, "we basically turn skin into a light bulb," said study co-writer Tom Kaye of the nonprofit Foundation for Scientific Advancement based in Sierra Vista, Arizona.

The findings were "helpful to learn about new non-destructive technologies that can help us study and document sensitive archaeological materials," such as mummies, said Deter-Wolf.

The oldest known tattoos ever discovered are on the remains of a Neolithic man who lived in the Italian Alps around five thousand years ago. Many mummies from ancient Egypt also have tattoos, as do remains from cultures around the world.

People have used tattooing for many reasons throughout history. The marks can show cultural or individual identity, social standing, and more. Tattoos also have been used for protection against disease or to deepen relationships with spirits or gods, suggested Lars Krutak. Krutak, who works at the Museum of International Folk Art in Santa Fe, New Mexico, was not involved in the research.

Designs on ancient textile and stone products are more commonly found in good condition for study than are ancient tattoos. But ancient tattoos when studied can provide information about art forms that would not otherwise be available, suggested Martin Smith of Bournemouth University. Smith was not involved in the recent study.

I'm John Russell.

Christina Larson reported on this story for the Associated Press. John Russell adapted it for VOA Learning English.

Vocabulary

mummy - n. ซากศพที่รักษาสภาพไว้ได้ดีมาก

visible - adj. ที่มองเห็นได้

exceptional - adj. ยอดเยี่ยม, โดดเด่น, ดีกว่าค่าเฉลี่ยมาก

glow - v. ส่องแสงเรืองรอง

light bulb - n. หลอดไฟ

textile - n. สิ่งทอ, ผ้า, เส้นใย

Reading Comprehension Quiz

1. How long have humans been creating permanent artistic designs on their bodies, known as tattoos?

For a few hundred years

For less than 1,000 years

For over 5,000 years

Since the Inca empire

2. What was the primary purpose of the recent study mentioned in the article?

To find new mummies in Peru.

To compare ancient tattoos with modern ones.

To uncover complex designs of ancient tattoos using lasers.

To identify the ink used in ancient Peruvian tattoos.

3. According to Michael Pittman, what made the fine details of the ancient tattoos hard to see?

The tattoos were covered by clothing.

The black ink used was too faint.

They were not visible to the human eye.

The mummies' skin was too fragile.

4. Who is Michael Pittman, according to the article?

The leader of the research team.

A study co-writer from a Chinese university.

An archaeologist from Peru.

A specialist in ancient ink technology.

5. Approximately how many mummies were examined by the researchers for this study?

About 775

Around 100

More than 5,000

Only four individuals

6. From which ancient culture were the mummies studied by the researchers?

The Inca empire

Coastal Peru's Chancay culture

Ancient Egypt

The Neolithic man's culture

7. Where were tattoos generally found on the individuals examined in the study?

Only on their faces.

Primarily on their legs and feet.

On the back of hands, knuckles, arms, or other body parts.

Exclusively on four specific individuals.

8. What made some of the tattoos "exceptional" as described by Pittman?

Their extremely old age.

They were found on royal mummies.

They featured complex geometric shapes like triangles and diamonds.

They were exceptionally well preserved.

9. How were the intricate tattoo designs on the mummified human from Peru's Chancay culture made visible for the provided image?

By traditional photography under visible light.

Using laser-stimulated fluorescence.

Both visible light and laser-stimulated fluorescence.

Through advanced digital rendering based on sketches.

10. What aspect of the ancient tattoos discussed in the article is currently not understood?

The quality of their artistic design.

The exact methods used for their creation.

Their resemblance to modern electric tattooing.

The individuals who created them.

11. What was Aaron Deter-Wolf's assessment of the quality of the ancient tattoos?

They were primitive and rudimentary compared to modern tattoos.

They were of a quality comparable to really good modern electric tattooing.

He found them historically interesting but artistically poor.

He suggested their quality couldn't be accurately determined.

12. In which publication did the results of this recent study appear?

The Associated Press

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Proceedings of the National Academy of Sciences

The Journal of Archaeological Research

13. How did study co-writer Tom Kaye describe the effect of using lasers to make skin glow?

Like turning skin into a mirror.

Like turning skin into a canvas.

Like turning skin into a light bulb.

Like turning skin into a textile.

14. According to Aaron Deter-Wolf, what is a significant advantage of the new non-destructive technologies used in the study?

They help uncover new mummies more quickly.

They allow for the creation of replica ancient tattoos.

They assist in studying and documenting sensitive archaeological materials.

They provide a way to date ancient tattoos more accurately.

15. Where were the oldest known tattoos ever discovered found?

On mummies from ancient Egypt.

On remains from cultures around the world.

On a Neolithic man in the Italian Alps.

In coastal Peru's Chancay culture.